



Notified, Not Yet Loaded

Predictive Ready-for-Pickup Alerts Against a Parcel Locker's Own Door-Sensor Confirmation

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Background

Notification speed is a published service target; notification accuracy is not. Facilities' predictive alert was rolled out to shave the SLA window. Whether it also changed how often students arrive to a locker that isn't loaded yet went unmeasured — until now.

Aims

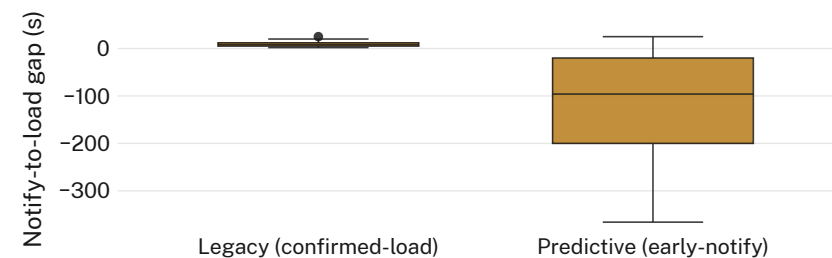
- Benchmark each system's notification timestamp against the locker's own door-sensor load event
- Quantify how often a notified pickup arrives to a locker that isn't loaded yet
- Test whether narrowing the predictive threshold recovers accuracy without losing the speed gain

Methods

- n = 214 locker assignments · 3 banks · 11 weeks (Feb–Apr 2026)
- door contact-sensor log as ground truth vs app push-notification timestamp
- 90-second predictive threshold held constant; pre-registered, no override

Results

The predictive threshold delivers on its speed target: median notify-to-load gap moves from +8s under the legacy system to -96s. The trade shows up on arrival — 18% of predictive pickups reach a locker not yet confirmed loaded, versus 0.9% under the legacy system (n = 214).



Predictive notifications precede the door sensor's confirmed load by a median of 96 seconds (range -6.1 to +0.4 min), against the legacy system's near-zero, non-negative gap, across 214 locker assignments over 11 weeks (Feb–Apr 2026).

Discussion & conclusions

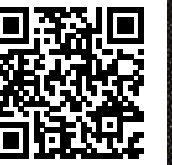
The predictive feature meets its notify-speed target by design: the alert fires at manifest scan, not at physical load, so the gain and the risk share one mechanism. The locker's own indicator light still tracks the sensor, not the notification, so the two increasingly disagree on the same door. Next: sequencing the push behind the light rather than ahead of the manifest.

References

1. Ranjbari, A., Diehl, C., Dalla Chiara, G., & Goodchild, A. (2023). Do parcel lockers reduce delivery times? Evidence from the field. *Transportation Research Part E: Logistics and Transportation Review*, 172, 103070. doi:10.1016/j.tre.2023.103070
2. Hoff, K. A., & Bashir, M. (2015). Trust in automation: Integrating empirical evidence on factors that influence trust. *Human Factors*, 57(3), 407–434. doi:10.1177/0018720814547570
3. Rittenberg, B. S. P., Holland, C. W., Barnhart, G. E., Gaudreau, S. M., & Neyedli, H. F. (2024). Trust with increasing and decreasing reliability. *Human Factors*, 66(12). doi:10.1177/00187208241228636
4. Jacoby, M., Tan, S. Y., Henze, G., & Sarkar, S. (2021). A high-fidelity residential building occupancy detection dataset. *Scientific Data*, 8, 280. doi:10.1038/s41597-021-01055-x
5. Marris, V., Tolan, A., & Vasseur, I. (2026). *Waved Through, Not Booked In: Hand-Signal Sequencing and the Loading Dock's Advance-Booking Forecast*. Slop University technical report. doi:10.5555/slop.zi0r6x

Deidentified locker-assignment logs only; no parcel contents recorded; predictive threshold pre-registered, held constant throughout.

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“The alert said it was ready. The sensor disagreed.”

18% of predictive pickups arrived before the locker actually loaded (n = 214, Feb–Apr 2026).